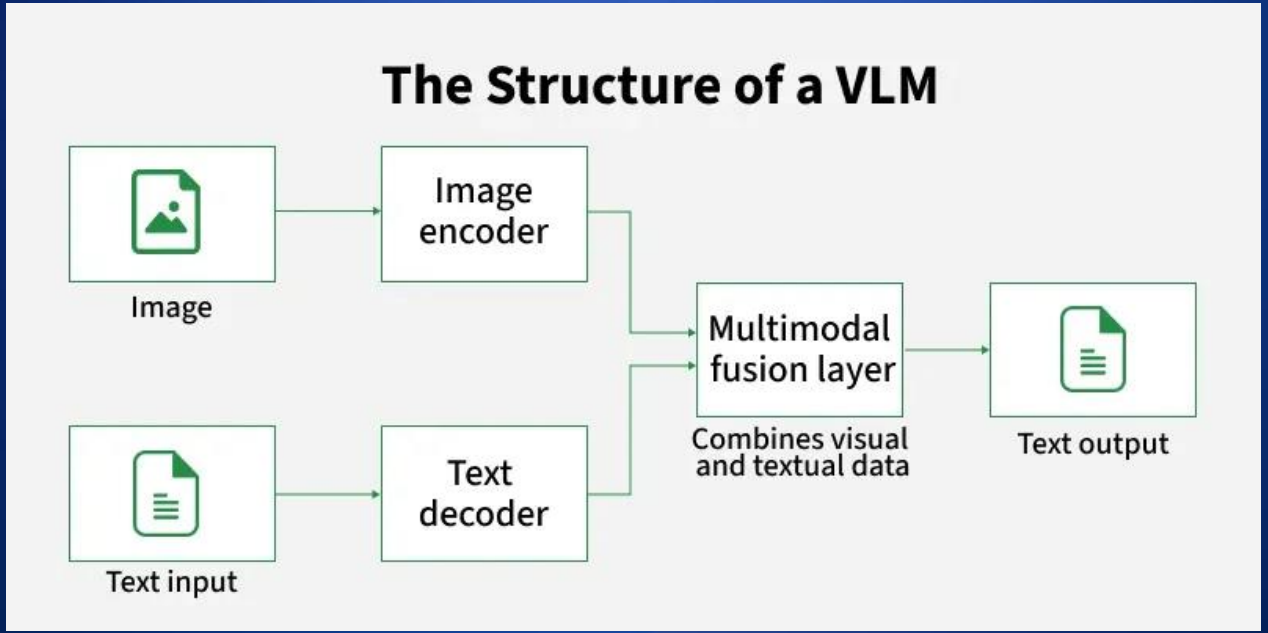
NeuroCogniAI

ARTIFICIAL INTELLIGENCE IN DEEP LEARNING / GENERATIVE / AGENTIC AI MODELS

VISION LANGUAGE MODEL (VLM)

Vision Language Models (VLMs) are AI systems that combine computer vision and natural language processing (NLP) to understand, interpret, and reason across both image/video and text inputs.



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TYPES OF VISION LANGUAGE MODEL (VLM)

Types of Vision Language Models (VLMs)

Vision-to-Text

Convert images into text

Text-to-Vision

Create images from text descriptions.

Cross-Modal Retrieval

Find images or text based on the other.

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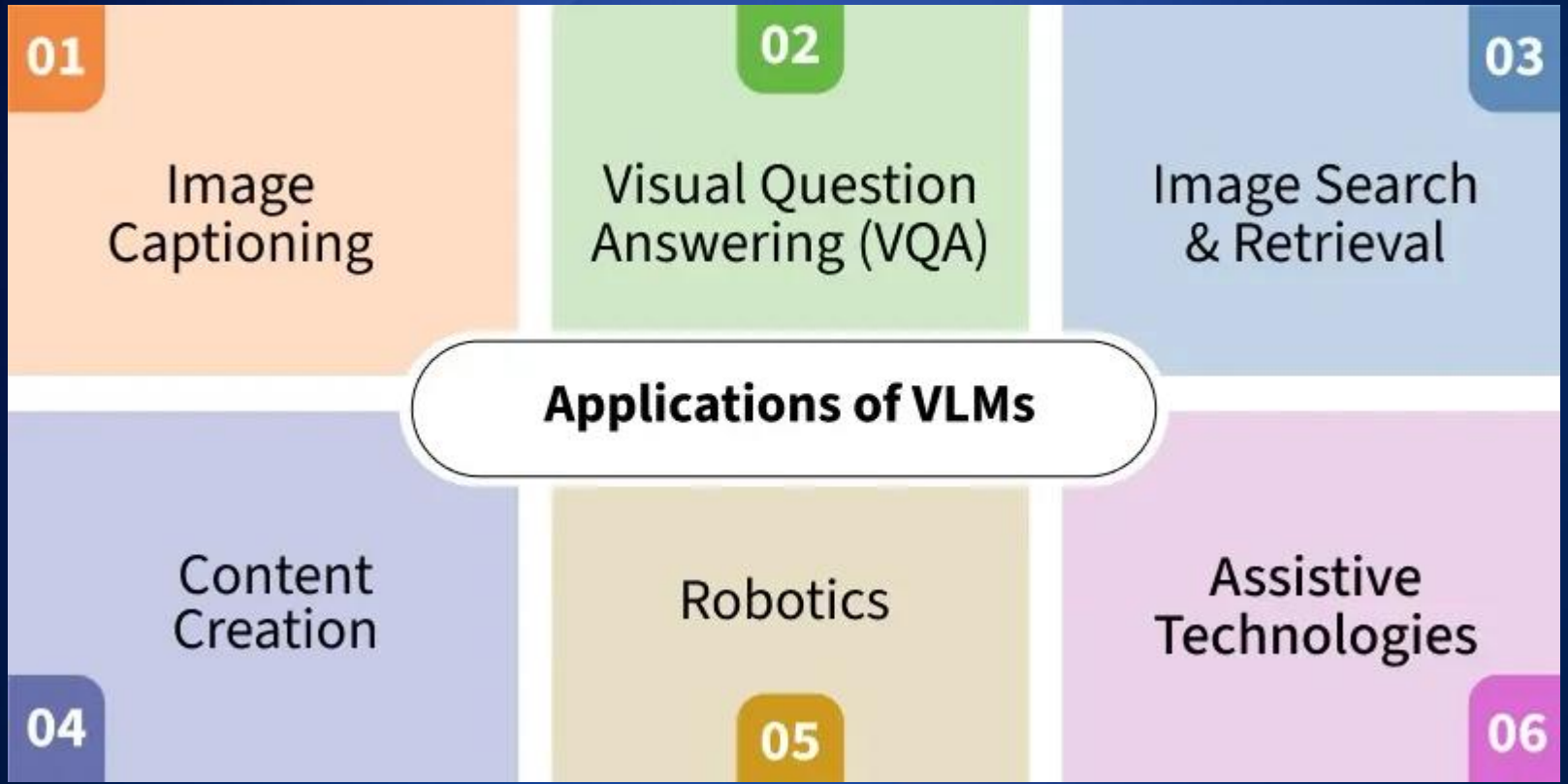
APPLICATIONS OF VISION LANGUAGE MODEL (VLM)



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COMPARISON BETWEEN LLM , VLM & VLA

Model Type	Processes	Examples	Use Cases
LLM (Large Language Model)	Text only	GPT-4, BERT, LLaMA	Chatbots, Content Generation, NLP
VLM (Vision-Language Model)	Images + Text	CLIP, BLIP, Flamingo	Image Captioning, Visual Q&A, Search
VLA (Vision-Language-Action)	Images + Text + Actions	PaLM-E, RT-2, SayCan	Robotics, AI Agents, Autonomous Systems

IMPLEMENTATION OF VISION LANGUAGE MODEL (VLM)

Step 1: Import Required Libraries

- We need PyTorch for tensor operations and model inference.
- PIL is used to open and process images.
- Transformers library provides the pre-trained Qwen2-VL model and processor.
- qwen_vl_utils contains helper functions for processing images for the model.

```
import torch
from PIL import Image
from transformers import Qwen2VLForConditionalGeneration, Qwen2VLProcessor
from qwen_vl_utils import process_vision_info
```

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IMPLEMENTATION OF VISION LANGUAGE MODEL (VLM)

Step 2: Load Pre-trained Model and Processor

- Specify the model ID for the Qwen2-VL model.
- Load the pre-trained Qwen2-VL model with bfloat16 for efficient GPU memory usage.
- Load the processor which will handle text and image preprocessing.

```
model_id = "Qwen/Qwen2-VL-7B-Instruct"
device = "cuda" if torch.cuda.is_available() else "cpu"

model = Qwen2VLForConditionalGeneration.from_pretrained(
    model_id,
    device_map="auto",
    torch_dtype=torch.bfloat16,
)

processor = Qwen2VLProcessor.from_pretrained(model_id)
```

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IMPLEMENTATION OF VISION LANGUAGE MODEL (VLM)

Step 3 : Generate Function

- Loads an image and prepares it with the user query for processing by a Vision-Language Model (VLM).
- Uses the processor to format text, image, and chat structure into model-ready inputs.
- Generates a response from the model using both visual and textual information.
- Decodes the generated output into readable text and returns it as the final answer.

```
def generate_answer(image_path, query, max_new_tokens=256):
    image = Image.open(image_path).convert("RGB")
    sample = {
        "messages": [
            {"role": "system", "content": [{"type": "text", "text": "You are a Vision Language Model. Answer concisely."}]},
            {"role": "user", "content": [{"type": "image", "image": image}, {"type": "text", "text": query}]}
        ]
    }

    text_input = processor.apply_chat_template(sample['messages'][1:2], tokenize=False, add_generation_prompt=True)
    image_inputs, _ = process_vision_info(sample['messages'])

    model_inputs = processor(text=[text_input], images=image_inputs, return_tensors="pt").to(device)

    generated_ids = model.generate(**model_inputs, max_new_tokens=max_new_tokens)

    trimmed_generated_ids = [out_ids[len(in_ids):] for in_ids, out_ids in zip(model_inputs.input_ids, generated_ids)]

    output_text = processor.batch_decode(trimmed_generated_ids, skip_special_tokens=True,
clean_up_tokenization_spaces=False)

    return output_text[0]
```

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IMPLEMENTATION OF VISION LANGUAGE MODEL (VLM)

Step 4: Running the Function

- Provide the image path and the text prompt you want the VLM to answer.
- The function is executed, and the generated response is printed.

Used image is:

Input image

```
image_path = "/content/image.png"
query = "Describe this Image"
answer = generate_answer(image_path, query)
print("Generated Answer:", answer)
```

Output:

Generated Answer: The image shows a young panda bear climbing a tree. The panda has a fluffy white body with black markings on its ears, eyes, and limbs.

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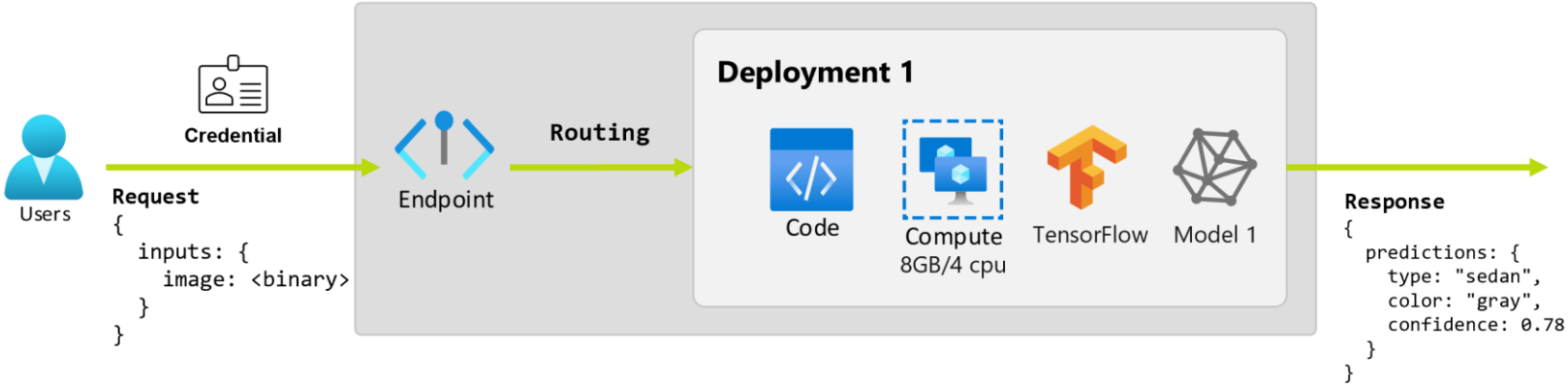
ML / DL UTILISATION USING ENDPOINTS



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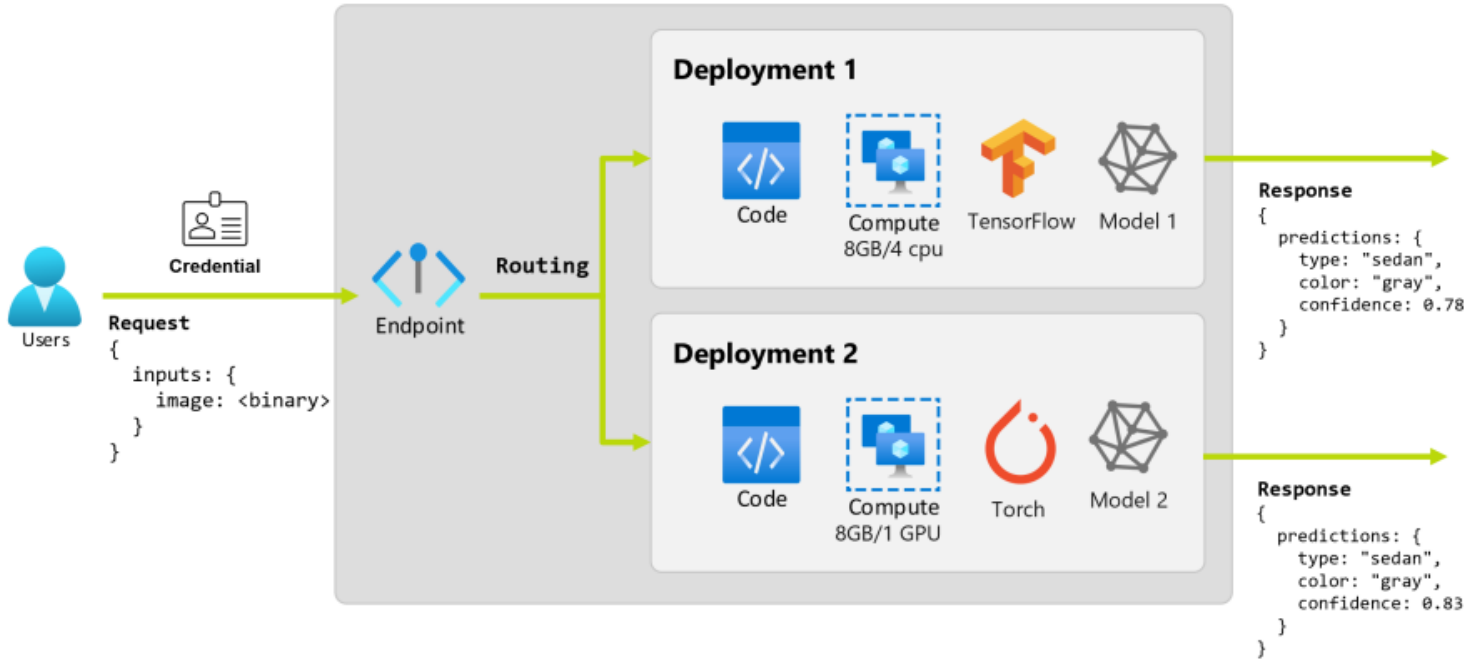
ML / DL UTILISATION USING ENDPOINTS



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